Manufacturing Operational Insights Report

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 4th Year 2026

Problem Statement & Objectives

1. Problem Statement

In today's competitive digital environment, creating an effective marketing strategy requires analyzing multiple factors such as target audience, competitors, budget allocation, and campaign channels. This process is often manual, time-consuming, and dependent on human expertise, which can lead to inefficiencies and inconsistencies in planning.

There is a need for an intelligent system that can automatically generate structured marketing plans from high-level objectives. The challenge is to design an AI-based solution that can understand user inputs, break them into actionable tasks, manage dependencies, and produce an optimized execution strategy.

2. Project Objectives

The main objectives of the *Marketing Planning Assistant Agent* are:

- To develop an AI-powered system that generates marketing plans from high-level inputs
- To implement multi-step reasoning for task decomposition and planning
- To integrate tool-based interactions for retrieving relevant data
- To generate structured and actionable execution strategies
- To design a user-friendly interface for input and output visualization
- To demonstrate the practical implementation of Agentic AI concepts

3. Scope of the Project

The scope of this project includes:

- Designing and developing a Planner Agent for marketing strategy generation
- Processing user inputs such as campaign context, stakeholders, and objectives
- Generating structured outputs including step-by-step plans and summaries

- Using mock tools and APIs for simulation of real-world data
- Implementing the system using Python, CrewAI/LangChain, and LLM APIs

The project does not include real-time integration with external marketing platforms or live data sources and focuses on demonstrating the planning and reasoning capabilities of an AI agent.

Proposed Solution

1. Key Features

- AI-powered marketing planning agent
- Converts high-level goals into structured plans
- Multi-step task decomposition and reasoning
- Tool integration for data retrieval (mock APIs)
- Generates detailed marketing strategies and timelines
- User-friendly interface for input and output
- Downloadable marketing plan reports
- Efficient session handling and response optimization

2. Overall Architecture / Workflow

The system follows an Agent-based architecture, where a Planner Agent acts as the central component responsible for generating marketing plans.

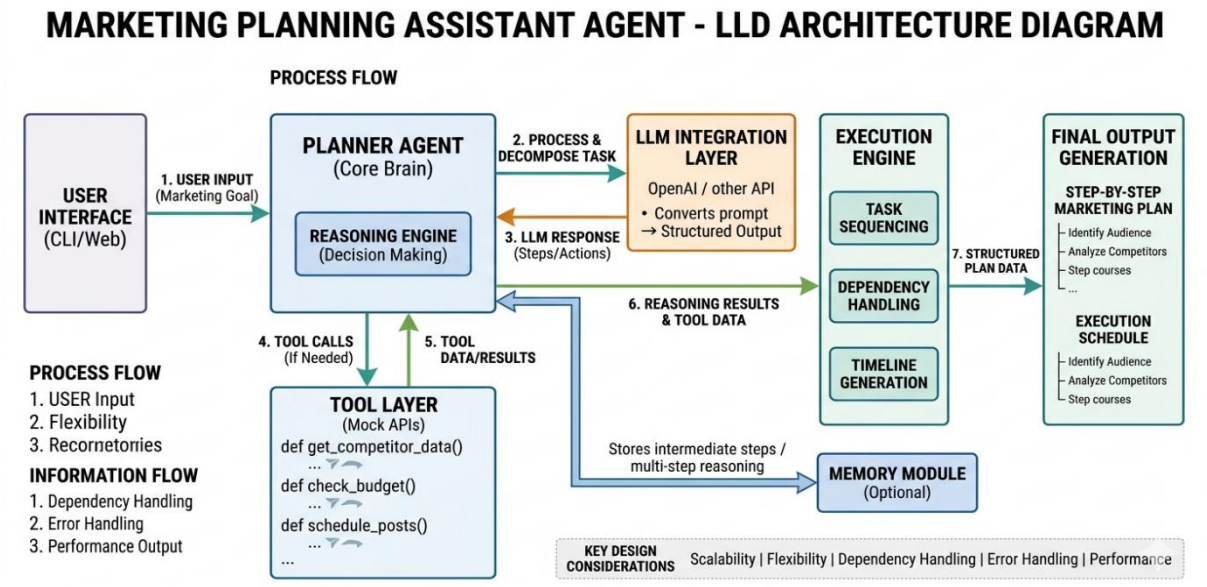

Fig. Architecture Design

Workflow:

1. The user enters campaign details such as stakeholders, marketing context, and campaign objectives through the interface.

2. The Planner Agent receives the input and forwards it to the LLM API.

3. The LLM processes the input and decomposes it into smaller actionable tasks.

4. The agent interacts with tool modules (mock APIs) to gather required information such as competitor insights or budget details.

5. The execution engine organizes tasks, manages dependencies, and creates a structured sequence.

6. The final marketing plan is generated and displayed to the user, with an option to download the output.

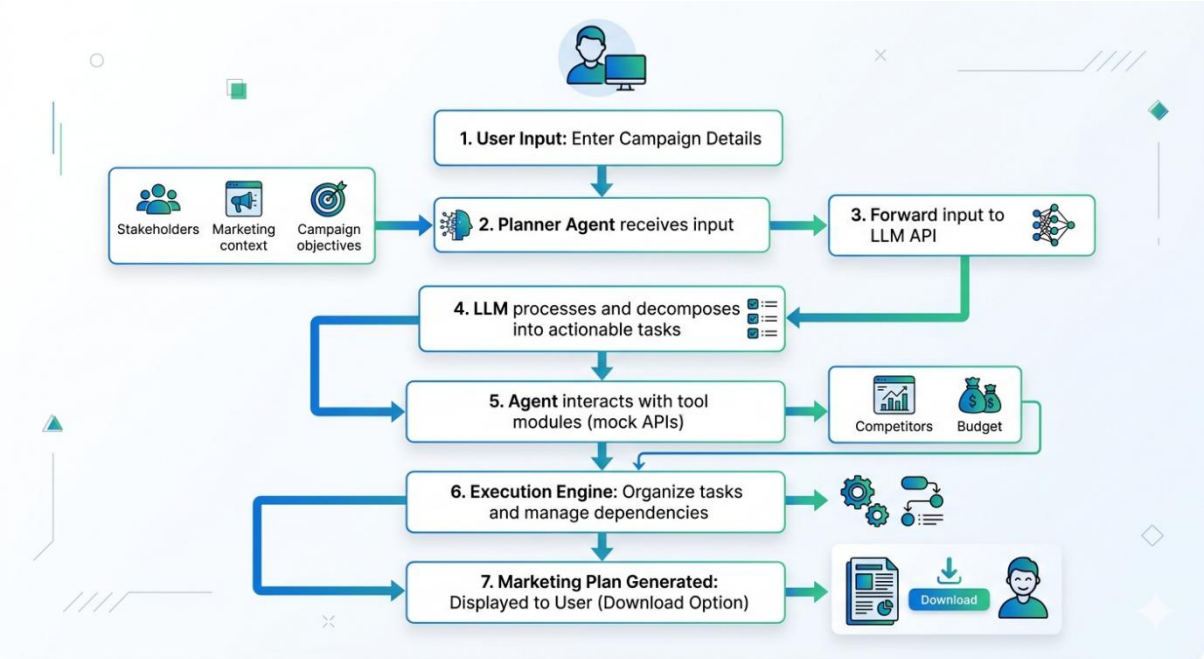

Fig. Flow Chart Diagram

State and Session Management

The system follows a stateless architecture, where each request is processed independently. A temporary session is created during execution to store intermediate data such as task breakdown and tool responses. After generating the final output, the session is cleared automatically to ensure efficient resource usage.

Caching Mechanism

To improve performance, the system uses a basic caching mechanism. Frequently used LLM responses and tool outputs are temporarily stored in memory to reduce repeated computations and API calls. The cache is short-lived and is cleared after the completion of each session.

3. Tools & Technologies Used

- Programming Language: Python
- Frontend: React.js
- Backend: Python (CrewAI / LangChain)
- AI Integration: OpenAI / LLM API (via OpenRouter)
- Libraries: CrewAI, LangChain
- Development Tools: VS Code, GitHub

Results & Output

1. Screenshots / Outputs

The system provides a user-friendly interface where users can input marketing campaign details and receive structured outputs.

- Input interface for entering campaign information such as stakeholders, marketing context, and objectives
- Generated marketing plan displayed in a structured format with actionable steps
- Step-by-step execution strategy including planning and scheduling
- Option to download the generated marketing plan as a report (TXT format)

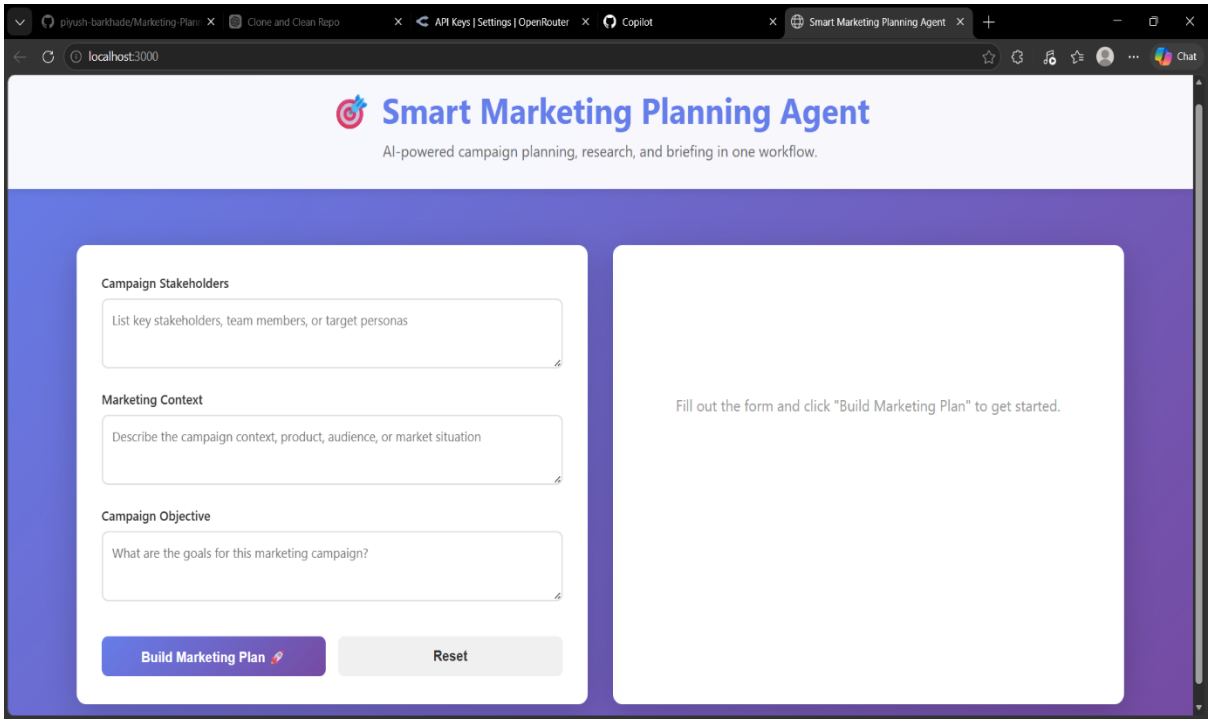

Fig. User Interface

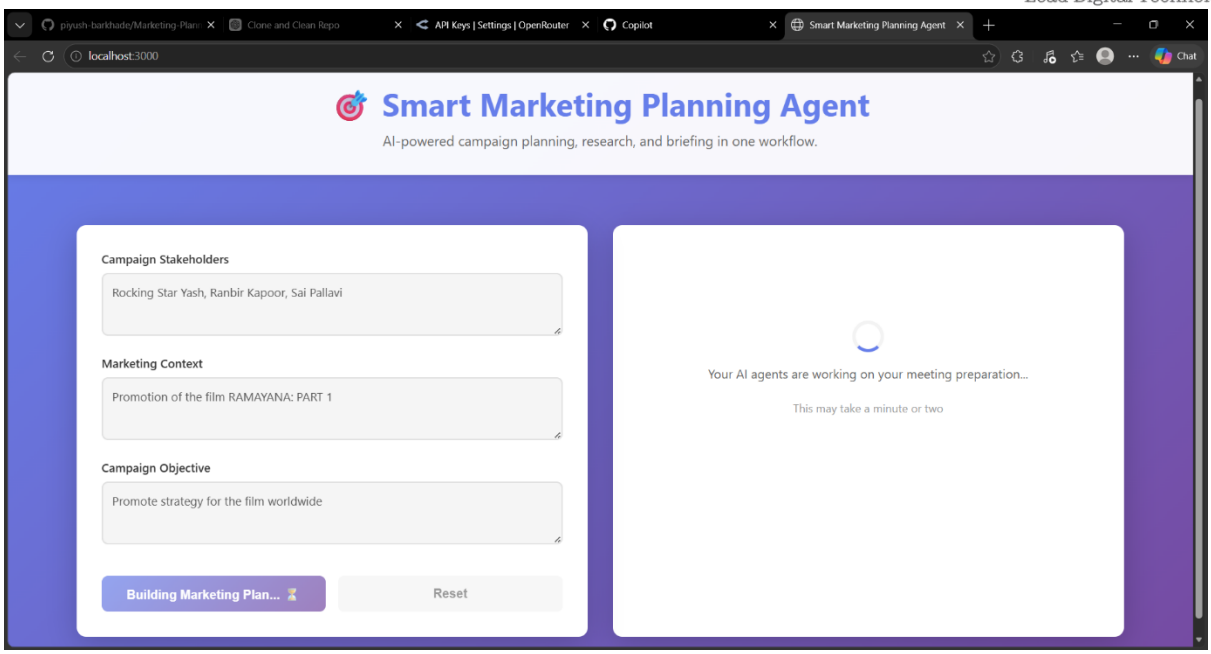

Fig. Form Fill

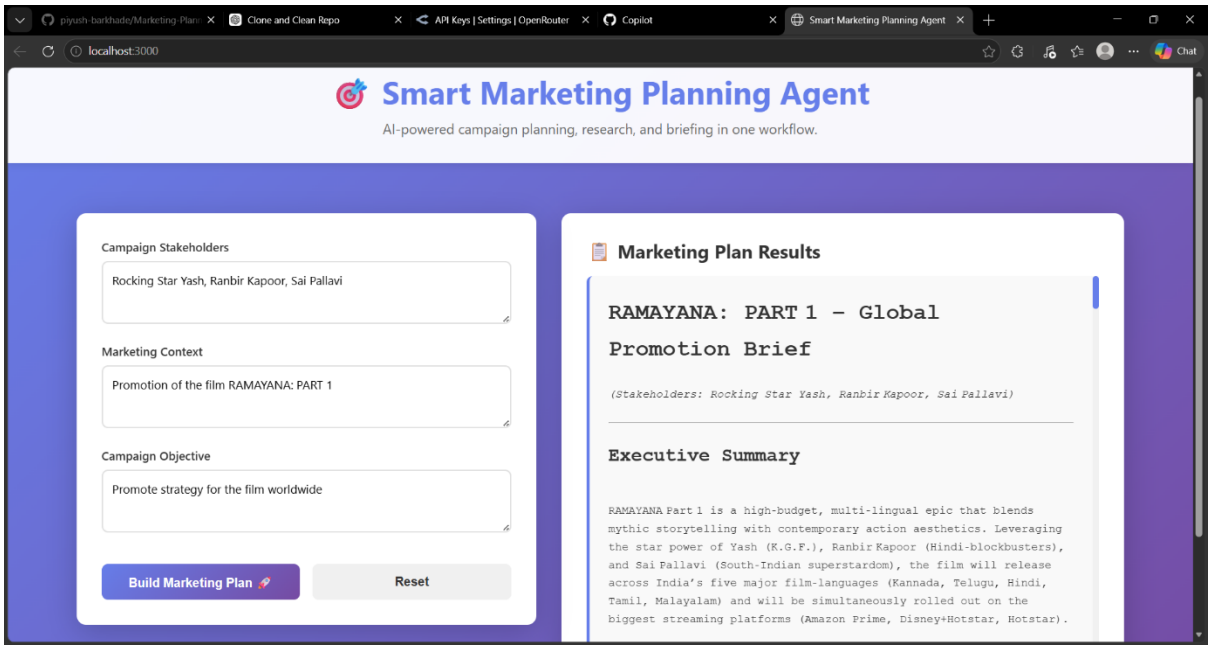

Fig. Result & Outcomes

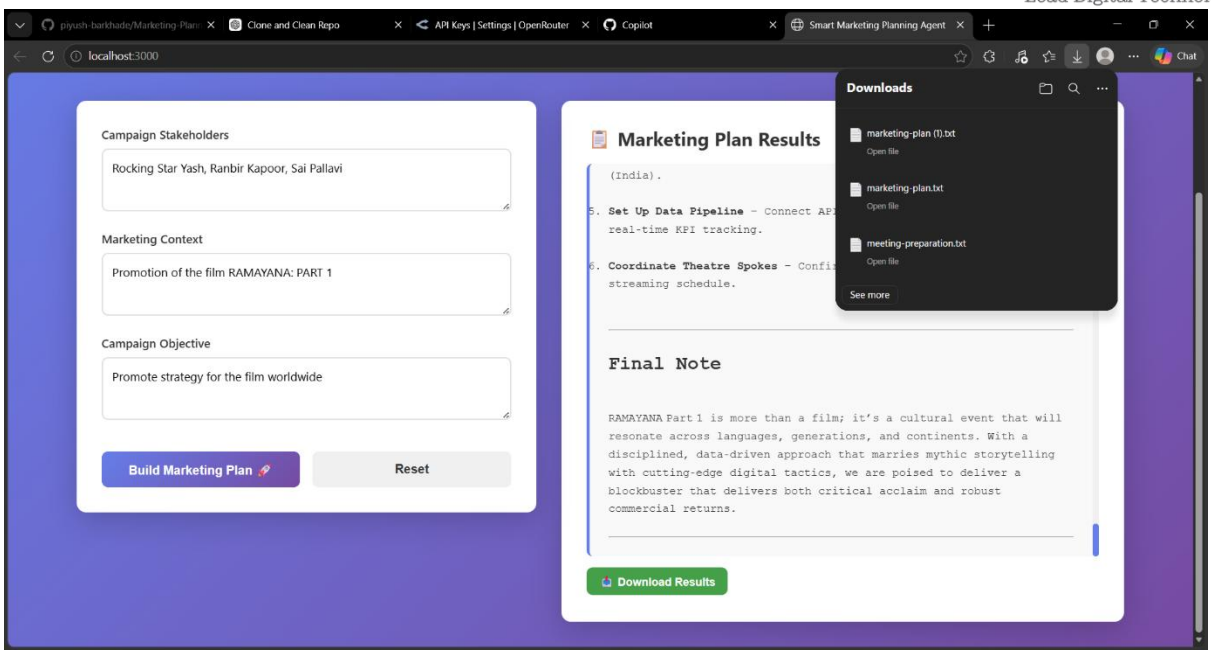

Fig. Downloading Txt

2. Reports / Dashboards / Models

- Generated Marketing Strategy Report based on user input
- Structured Execution Plan with task decomposition and sequencing
- Campaign planning output including audience targeting, competitor analysis, and budget allocation
- AI-generated content summaries and recommendations for marketing activities

3. Key Outcomes

- Successfully developed an Agentic AI-based marketing planning system
- Automated the process of converting high-level goals into structured plans
- Implemented multi-step reasoning and task decomposition
- Demonstrated integration of LLM APIs with a software application
- Improved efficiency and reduced manual effort in marketing planning
- Gained practical experience in designing intelligent agent-based systems

Conclusion

The Marketing Planning Assistant Agent successfully demonstrates the application of Agentic AI in automating complex marketing planning tasks. The system is capable of transforming high-level marketing goals into structured and actionable plans using multi-step reasoning, task decomposition, and tool integration.

Through this project, we implemented an intelligent agent that simulates human-like decision-making and planning capabilities. The system improves efficiency by reducing manual effort and providing organized execution strategies.

This project also provided valuable learning in areas such as AI integration using LLM APIs, agent-based system design, workflow automation, and building user-centric applications. Overall, it highlights the potential of Agentic AI in solving real-world problems and enhancing productivity in the marketing domain.

Future Scope & Enhancements

The *Marketing Planning Assistant Agent* can be further improved and expanded in several ways to enhance its functionality and real-world applicability:

- Integration with real-time marketing platforms such as Google Ads, Meta Ads, and analytics tools
- Addition of a database system to store user history, sessions, and past marketing plans
- Implementation of advanced analytics and performance tracking for campaigns
- Development of a multi-agent system where different agents handle specific tasks like content creation, budgeting, and analysis
- Deployment of the system as a scalable web application using cloud platforms
- Enhancement of the user interface for better usability and interactive experience
- Support for multiple industries and customizable marketing strategies
- Incorporation of real-time data sources for more accurate and dynamic planning